ActInfLab 2022 Quarterly Roundtable #1

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[Music] hey everyone thanks for tuning in and today it is march 30th 2022. we're here in the actin flab quarterly roundtable number one for 2022. welcome to the actin flap we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at some of the links here on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following video etiquette for live streams today in the actin flap quarterly round table number one for 22 we're going to have a first section after our intros on lab scale updates and then we'll talk about project scale updates which are nested within the lab scale and just close with some preferences and expectations and through and through we hope that you'll see many affordances for participation many extensive cues signals in your epistemic niche that elicit action that make you curious about something that um connect with your pre-existing preferences or update them so that you get involved with active inference or possibly even the lab so whenever you do see this or listen to it or some other combination feel free to go to activeinference.org or email us at activeinference gmail.com and we'll be in touch from there so we can begin with just some introductions and warm-ups and it feels so different to do this without a paper that we've read before so for these intros we can just say hi and anyone can reflect on just like how is your generative model updated this quarter what did you perceive or learn how did you change and what might be something that you're looking forward to as we carry on the latter of 75 percent of 2022. so i'm daniel and a researcher in california i think my generative model updated and fragmented into a thousand little pieces with coda and the way that we used formal documents this quarter as opposed to previous quarters but it's cohering in a way that is actually helpful i hope i'll pass to alex yes everyone i'm alex watkin i'm one of the founders of the lab and also researcher at systems management school and from latest time i feel like we're sharing our generative model more and more and hopefully it will bring us to me to two new developments and they paste it to blue hi everyone i'm blue i am an independent research consultant in new mexico um i think this quarter was pretty special like the coda restructuring was pretty awesome and just contributing to building like lab infrastructure in coda and in like youtube with like videos and time stamps and podcasts and i don't know it was it was fun um i think probably the biggest update was was the quantum paper because that was pretty pretty epic um the one that we just have discussed over the last couple of weeks um and in the next quarter i'm looking forward to having some more quiet time less talking more viewing that's what i'm looking forward to in q2 of 2022. and i will pass it to yvonne thanks blue yeah because he's very nice i i'm i'm a violinist in uh many now and we finally run system thinking study group this uh this water and i hope lab participants enjoyed and i see it that they do and i hope we run next quarter another study group and it will be on like conference so that it is hi yeah a lot going on um i'm becoming a little less guilty about skipping at least half of the activities but i get to hear at least at least to read the papers and listen to the live streams and there's lots of work going on with the transcript which is very gratifying even if daniel no longer believes himself about how valuable it is to do the print transcripts and put them in in service i still am sticking with the 10 000 times more value from a transcribed talk than just a youtube audio with pictures so who haven't we heard from stephen hello yeah we're still here we're still standing sort of so that's great i'm here in toronto um for me i'm appreciative for certain of the capacity the labs offered the connections the conversations the ability to bounce the ideas and that's been really really helpful and um particularly with like mark's uh sones and a lot of these ideas with quantum which at first i thought some might be quite abstract we're starting to synthesize and find ways to to pull them together and so i'm hoping that can be used now i'm working with some embodied workshops and taking that into communities and letting this work inform how i coach how i facilitate how i try and explain that's the hard bit but uh i wasn't that and it could but shouldn't go without saying just massive appreciation to everybody who does and continues to show up for active lab you see the tip of the iceberg on the live streams but there are many others as we continue to formalize and determine and reduce our uncertainty about what kind of thing we are here we'll hopefully have clearer and clearer ways to source attribution and to show recognition for all the people who contribute in just the million ways that are possible from beginner to very familiar so we will just provide a brief overview of active lab from a narrative perspective before we jump into our strategy discussion so in 2020 towards the middle and in the ends some of us wrote a paper in uh september was when it was published on active inference and behavior engineering for teams which was addressing a gap that we perceived in the literature which was the relationship between active inference framework and remote teams and communication at the end of that year after thinking about what our next steps would be after that paper we thought that something like an active lab would be cool so we sent out an initial call for collaboration during all of 2021 was our first year of active lab project operation and the projects were in the three organizational units of education communication and tools this year in 2022 which is what we're going to be talking about today a lot has happened at both the lab scale and the project scale so we're going to first talk about the lab scale and the two pieces under lab scale updates are going to be the initial release of the lab strategy document and we'll also discuss the initial advisory board cohort and then we'll go into the project scale and discuss some of the ongoing projects that have resulted in all kinds of deliverables and outputs so first to the lab scale alex would you like to talk a little bit about just from a high level strategy yeah thanks i think as everything in our lab we are trying to do things in some new ways that was not using before and also trying to put it on some state-of-the-art terrace and approaches so as for strategy we consider strategy no not as a concrete document but as a ongoing process which will be which will have a quarterly reviews and updates and uh on high level we are following open handedness approach um for for who is interested it's now can be put can be approached from developments of canada stanley and we're trying to avoid using classic objective missions goals and other corporate stuff so we are trying to see in the focus of the future we are trying to see some stepping stones that we can make next steps and also for sure as from the beginning we use active inference not only as a research field but also trying to implement and use concepts from where such as preferences and expectations and also we consider that all our activities are on the different scales so we need to follow and understand how it's applied on different scales so thank you so we will now go through the current state of the strategy document and just anyone here feel free to raise your hand and of course anyone watching along feel free to write questions and we'll get to them pause the video or check it out to read the full text here but just to summarize our strategic document preamble in active lab we're directing our attention to different phenomena across multiple scales and the scales that we're most interested in or the ones that we are heuristically using are the personal the individual human scale the lab scale that's active lab scale and then the community or the niche or the environment scale which is the epistemic surroundings of the lab and what's special about the lab scale is that is where the lab can implement policy we can emit actions that are perceived as sense states by individual people or by the external community and that may or may not change their policy selection but the lab scale is where strategy happens because that's where we can actually implement policy selection and we're taking this multiscale nested systems perspective to five different aspects of actinflab and those five aspects that we're going to talk about in the next few slides are education research outreach and engagement methods and open-endedness any general comments before we continue okay so actinflab is an educational nonprofit this plays out at all scales as all things do but we're especially interested here in the personal scale in educating individuals in having individuals that learn and update their generative model and also respecting just the tremendous diversity of active inference journeys that different individuals are coming in and through relationship with the active inference lab and at the more community scale our educational efforts are oriented towards reducing research debt and increasing the applicability and the comprehensibility and the rigor and accessibility of active inference and related ideas any comments on education okay the second aspect of the lab is the research aspect and here it also plays out at the personal and at the community scale at the personal scale we hope to provide a niche that scaffolds and catalyzes the research projects by participants which is something we'll come back to later but we both do research in the lab as well as on the interface and just hope to motivate and create a space for people to share their individual research updates and at the community scale we're recognizing that research is carried out alongside ongoing discussions about responsibility and ownership and many many other terms so there we hope to learn practices and improve and give our own twist on what the future of science open science decentralized science dsi just science with no adjectives and education what will research and education be in our future stephen thanks daniel yeah this is really helpful to to read and also the fact that we've got lots of people from different backgrounds coming together we we're able the research that's happening is is quite trans-disciplinary and um or it's been informed by lots of different ideas so there's a kind of a mixture of those processes so it's helpful to have the systems engineering work that we'll probably come to later but that's helpful in letting us get a feel for how the research doesn't get siloed as purely a paper but okay how could that be bounced off to maybe make a difference in the world in ways that maybe people who don't even know active influence find that they're using something differently because active imprints has helped the way that their practice is shaped so i like that awesome the third aspect is outreach and engagement and the active inference lab is working on a global scale with a participatory model so at the personal scale we welcome all participation and engagement and that means all time zones and spoken languages and preferences and what people want to do and where they're at with active inference participation in a sense begins with where the participant is and there's of course more that i hope everybody can say and add and show not tell or show antel if they've brought enough for the class in the coming years and then at the community scale we also seek to be part of a emerging landscape of high quality science communication and participation and inclusion in research and projects around topics like active inference which is what we're most attuned to in our regime of attention but that doesn't mean we only talk to active inference people so we hope that our outreach and engagement is meaningful and sustained and impactful any comments on outreach and engagement methods the active inference lab applies state-of-the-art methods across domains that refers to the methodology of active inference and the emerging tools and ways that we have of working with active inference as well as other methodologies like for project management for education for communication etc so at the personal scale we're educating participants so that they can grow their expertise in different disciplines and also develop as a person holistically like keeping it balanced you're not going to stay up and finish reading every single active paper staying balanced and having methodologies that help us be performant but also balanced are quite important and then at the lab scale we hope to implement the cutting edge the pareto optimal the bayes optimal methodology which includes our switching strategies we fix typos on the fly that's what editable documents allow our affordances as an organization are the methodologies that we scaffold in projects so especially at the lab scale we hope that we use methodology that's powerful and facilitates collective intelligence and also clarifies the distinction or the delineation the partition perhaps even between those who are following along and listening to live streams reading the papers in the game that's awesome and then those who actually onboard into the lab and eventually have a role in a project will be able to increasingly be clear about what that means through the uses of different systems ideas like practices cases responsibilities and concerns yes yvonne yeah so thank you i would like to add that if we compile uh methods and outreach so the lab provides a lot of opportunities for the understanding of active fingerprints and we provide a lot of tools and for those who already participate in in the lab and but those who just start to deep inside the active influence the last but not really the last aspect is open-endedness as alex mentioned earlier and has been bringing and reminding us all of since the beginning that active lab is on the path of it doesn't mean that we sit and wait where the road forks or when we have uncertainty about the future those are fundamentals and in fact open-endedness is action amidst uncertainty so at the personal scale we recognize and support the open-endedness of the personal journey of individuals who are learning and applying active inference so even if you don't know exactly where you're headed or what your finish line or your milestones your checkpoint looks like if you don't even know what your preferences are or you're unsure about some other parameter in your generative model that's all good at the lab scale our strategy is to say it is a work in progress would be to almost assume that there would be a final strategy but rather our strategy is an ongoing process that will always be updating in response to changes in the niche and the effect that our actions have over multiple time skills we do recognize and utilize approaches to achieve short-term success and realizing products and deliverables over short terms but also there's a time and a space and a place for thinking about active inference and the active inference lab in deeper time and that is especially where diverse participation and our lab advisory board come into play which we'll talk about in a second and then just lastly at the community scale it's true in principle that active lab can participate in various kinds of activities that might be beyond education research and we will always be considering those opportunities in a case-by-case way and assessing whether it's something that fits best within the markov blanket and the kind of thing that active lab is or maybe some thing else could exist that could play that role in the ecosystem yes dave okay stephen i'm interested by that way that the ecosystem this this question of what's in the markov blanket and the non-equilibrium steady state and what's in the niche and we um those three things are kind of dynamic i think over time looking at the way the world is and without getting too evangelical there's lots of big you know there's big uh claims made about active inference and its potential as we've been having conversations we've seen a lot of things at scale come up which is quite exciting conversations around that but often maybe a little bit fuzzy a little bit conceptual so i think some of the stuff that's going to come out there the lab's going to be spinning off as much as it might be holding on to excellent thank you so that concludes the discussion on the strategy update the second lab scale update is to mention our advisory board so this information can be found at our website activeinference.org and on the top right head over to where it says the advisory board is a insight and guidance board it's not a managerial board and it exists to help the lab make better strategic decisions so we engage our advisory board participants in terms of their expertise different recommendations and questions they have for the lab and we recognize and hope to draw on and synergize with their experience in many many different areas we have an image here from kirchoff at all 2018 about nested and about the way that as systems develop for example as potentially a um over developmental time a zygote goes from being a single cell enclosed by a lipid bilayer towards increasingly ramified levels of embryonic complexity or similar transitions could be happening over evolutionary time something like that is occurring as we elaborate and build the active lab we have an awesome initial cohort of the advisory board and it includes bradley alicia john boyk matt brown john clippinger scott david jeff emmett chris fields carl fristen raphael kaufman anatoly levanchuk rosalyn moren alba serrano cheryl van hoof tim verbellen swann webb and michael zargham and we have already received really important and meaningful feedback from them and really appreciate it and for those who are interested in the kinds of work that we're up to and might have experience in a certain domain that could be helpful for example related to educational non-profits then perhaps get in touch with us but this is a major change to the lab scale to introduce an advisory board and we're off to a fun start with them so far so we look forward to just seeing how that continues any other comments on active at the lab scale or we can head to the projects okay so as we head through these projects feel free to raise your hand to any point if you want to add a comment or a question and as a participatory lab those who are listening and you've made it this far recognize that all of these reflect the contributions of participants and reflect affordances for you to contribute further even if only to reduce your uncertainty about the documentation or the way that something is occurring so hopefully if you're listening to this string of symbols know that you can participate and perhaps you even should we're going to talk about a few different projects today in fact 11 difference of them and you can see loosely on the left that these are in three organizational units edu comms and tools but the project scale is really where the team and the roles and the practices occur so we're highlighting the project scale and the project is also the unit of granularity where you can really get involved and make a contribution so showing up for our synchronous organizational unit meetings is an awesome way to kind of paint a first coat and to be exposed to many of the projects but by and large the contributions happen at the project scale and asynchronously so all backgrounds all time zones all levels of commitment everyone is welcome to get involved with these projects which we're going to go through now they're going to be the active inference ontology the active inference course internal educational initiatives like the systems thinking reading discussion group and ontology working group active inference journal research updates live streams podcasts social media active block reference robotics and embodied and active lab meta modeling on we go to the ontology we've just recently pushed an update to the active inference ontology and it's available at this coda link here the goal of the active inference ontology is to increase the accessibility and impact of active inference by increasing clarity and interoperability and expressivity and all these other good things about terms and ideas and when you go to this site you'll find that there's a explanatory preamble and what this ontology release contains primarily are two different kinds of things proposed definitions and translations so first the proposed definitions this is what the proposed definitions look like for the core terms the terms are on the left column here and then everywhere there's an at and a blue is like a clickable link to another term so it's kind of like how dictionaries use other words in that language and it will say a is kind of like b except it's different than c and so that's what the definitions are they're natural language answers to the question what is x and so these proposed definitions are currently in natural language but there are other ways to develop them more formally the proposed definitions are there for those who just say well what does this term mean it also helps with computationally aided sense making and then the second piece that is in this current release of the is the translations and so here's the current view of the top of the translations table so we have the term in the english on the left side and then we have so far complete translations of the core ontology terms into spanish portuguese french italian czech russian and hebrew and some german and some others that are sort of beginning so especially if you have fluency or friends with fluency in some of these languages but especially new and different languages this is a really awesome opportunity to translate some terms which are mostly normal natural language terms into another human natural language and that will facilitate the ability to for example deploy educational information or transcripts in their translations do it in a way that's rigorous and accessible and meets people where they are in terms of their language fluency that's the active ontology um project where we're developing a participatory um ontology for active inference terms could i just ask not putting you on the spot but if i was to ask you like what what areas have been the most challenging or the most um kind of uh like uh most most difficult to overcome and um what what are the sort of questions that are still being grappled with on the ontology you know and which ones have we overcome already and which ones are still sitting there in the background i'm just curious if you've got any thoughts on that yeah good question i'll give it a thought and then anyone else is welcome to if they want one interesting thing is these definitions although many papers do sometimes define their terms either in the text or in a box or in a glossary that doesn't mean that the way that the terms are defined are coherent or compatible or consistent and a lot of times it's like a yes and someone defines it one way and someone else just kind of takes a different spin on it but other times we have come across several situations all together all of us at least here where there were incompatible seeming uses of terms and perhaps that happens with all kinds of linguistic elements so that was one interesting thing was that it's not just like we're looking through papers and copying sentences there's actually a really powerful process of the discussion that we had focusing week by week on just a few terms or one term where we were seeing all the different ways that different terms are being used inside and outside of active inference and just trying to work towards some kind of grip on that blue one of the biggest challenges that i think we face um is dirty and incomplete data which is like pervasive across fields and disciplines everywhere so i mean trying to be the bridge to fill the gaps to make the data to complete the data like and i mean there's always opportunities that people want to get involved like in filling out the data like helping us to complete um you know to have a complete uh picture of which to metamodel awesome and there are other columns and features of the ontology that we're working on in the lab for anybody who wants to get involved like correct examples just simple declarative concise positive examples of terms being used correctly that's awesome incorrect examples people who are using sentences it doesn't have to be caught in the wild but it could be but just generating sentences that are like counter examples can be very helpful for both humans and computers in their learning and so we have examples um counter examples formalisms connecting core terms and other terms to equations and papers and connecting it with the literature corpus so there's so many angles those with curiosity and expertise in active inference and or ontology should absolutely get involved with this which will be a long-term project of the lab and we hope to find new ways of deploying it and just to give one little contextualizing slide here this entire continuum can be considered to be different types or grades or like a taxonomy of ontologies and so ontologies in their most informal can be as simple as a list of terms or glossaries the sources or what we have right now with the current release which is like a principled informal hierarchy and some relationships amongst terms and that's what we have right now with the proposed definitions and translations we're also working with the ontology working group which we'll talk about in a few minutes on formalizing that ontology for example using sumo and suo kif and thinking about how do we integrate the ontology with course development with knowledge engineering and literature integration so some of the many things that people can look forward to on to the second project which is the active course in the active course project we are seeking to create a learning platform that embodies the principles of active inference and facilitates the learning of its principles and concepts to learners of many different backgrounds and this is something that we are not uh too far down the road on but are eagerly looking forward to especially in light of some recent epistemic changes in our niche which we'll talk about soon but there's a massive potential for active inference education and we hope to have course offerings and educational experiences that meet that opportunity alex yeah just a quick note to avoid possible misunderstanding that notion platform here is not like for some it platform but more about a set of educational services which will be supported to improve learning affordances yes good call thanks for clarifying that if you're interested or curious about course design education education in a participatory way non-profits those are all excellent cues for people who would like to come on board and help in a serious way for this course as we start to think about what it can be third project we have several internal educational initiatives in active lab and so on the left we see the systems thinking reading discussion group and on the right ontology working group the owg alex or anyone would you like to describe or give a note on either of these sides i just want to say that from very beginning we consider that we should not only learn and educate active inference but also to learn as you mentioned before state-of-the-art practices and system thinking was one of the main discipline from the beginning what we think is very useful for collective work especially in remote mode so it's great but now more and more people start learning and it definitely will improve our communication and speed up our activities on the projects thank you yes yvonne has been facilitating the systems thinking reading discussion group strdg and we have an intrepid group of participants who are working their way through this online course and having regular discussions and asynchronous conversations as well around it and as alex said this is really important for the ways that we work together and communicate and on the right side we have the ontology working group so previously we were discussing the active ontology as a project where we're talking about active inference terms and how they're related in the ontology working group we are increasing our competency in computational ontology and what that has looked like primarily has been working with the sumo suggested upper merged ontology system in suo kif and working through the textbook of adam peace and this has been an awesome experience dave has made a massive amount of contributions and ontological um limericks of different formats and that has also been a great learning experience that hopefully will fold back in towards improving the active inference ontology but it's also just a learning initiative and that brings us to a ongoing learning initiative which is the release literally in the last several days of the textbook by thomas parr giovanni pizzulo and carl firsten it's the active inference textbook and we asked well what can the active lab do for slash with the textbook do you want to join a textbook group not just a reading group just a textbook group because we are going to be organizing cohorts of textbook groups and you can email us for the form if you don't already have access to it but you will fill out a form that asks whether you'd like to join an upcoming textbook group cohort and it will be done in groups in cohorts yes you're ready or no you'd like to perhaps hear about that affordance in a few months we ask for name email address and country whether you have a textbook or not and then just for you to provide several sentences or more on your level of familiarity and your interest in active inference and all backgrounds and familiarities as with all activities are welcome this is going to be a really experience i think to work through this textbook and see it as just one pebble on the ant hill or the tip of the iceberg or any other metaphor that people use for the way in which this textbook will both be a visible and centralizing artifact as well as something that is really only a few hundred pages long and it's a relatively small physical book it's just the beginning it's not the end of active inference and those in the textbook groups will realize that and enact it any comments on internal educational initiatives alright number four the active inference journal so in this project we are publishing knowledge artifacts so that they are searchable indexable and citable most recently in the active inference journal we uploaded a second improved version of the transcript of the carl fristen symposium applied active inference symposium that was in june 2021 and from that we are improving our methodology that we learned and learned by doing in that first symposium transcript process and applying that to other event recordings for example guest stream 16 with mark solms and live stream number 40 with a quantum free energy principle so um the journal is going to be a great place where we continue to build a participatory pipeline for improving and written dissemination anything that anyone wants to add on this yeah i sent the uh the original paper and the three uh video segments to sean carroll along with a summary of the three big points in the summary he says this is cool i will check it out uh which i'm sure incentivizes the gentleman who volunteered to go through our transcript and make it more presentable to get through that and um then we can send that to the very busy doctor carol who just has a new appointment he is something like the chair of the philosophy of science at johns hopkins all his pain of being denied tenure is paying off apparently and he may pay off he already i think gets along very well with professor tristan he uh he says i have never had to work so hard in an interview with somebody who is so knowledgeable and well spoken and anxious to communicate yeah and it's a great example of a project increases accessibility for example by taking um two or six hours of youtube discussions and making them searchable and translatable and then also reduces research research debt because we can have somebody spend three or seven hours improving and curating a manuscript and finding oh when somebody mentioned this paper here's that specific citation that multiplied by thousands of people searching for that paper maybe they have access maybe they don't maybe they misheard the name there's so much that we can be proactive about in terms of making our knowledge corpus really accessible and meaningful and woven according to modern practices awesome work though dave and others on the journal maria and others as well on um five we have research updates so this is not a project that we actively steward per se in the lab but this is just capturing research that's in progress and finished by lab participants so just to highlight a few papers and products that have been released by active inference lab participants but not necessarily as an active lab project in the last few months so by avail gwen and carlu aka serval aka icarus uh the paper cognitive agency in sociocultural evolution and then a awesome collaborative paper that we wrote with myself shawn arhan rj shadi avel blue yvonne sid ahmed yaakov caleb and alex where we talked about decentralized science desai we integrated some of the more technological perspectives on dsi for example related to web 3 and blockchain and juxtaposed that with some of the bottom up the view from the bottom participatory sense-making and epistemic commons and produced an active entity ontology for science eos that we hope will allow us have a way to model some of these complex epistemic ecosystems in the coming months so that's just some of the research and there's other awesome research that's in progress on to yes stephen go for it yeah be curious i think this uh work with situated sense making for decentralized science is really really interesting um and i think there'd be ways that the other areas of sense making which this can be uh helpful to bridge that active generative modeling approach with uh other ways so i'm i'm really glad that you did this work actually as a group and uh i think it's gonna help tell another story which might have been a bit too hard to tell from the ground up but now you've got a bridge so thanks okay on to six we had live streams as usual in this quarter at the coda dot io active inference lab active lab you can find the past and upcoming live streams uh several guest streams and other kinds of streams but most of our streams in the last few months were the paper discussions so in live stream number 35.01 and two so as with all of these there was a zero a background and contextualizing video that was prepared in the weeks leading up to the dot 1.2 discussions which are participatory group discussions so if you ever see a paper where you want to be in the dot 1 or the dot 2 and be part of a group and you don't need to say a ton no one's gonna call on you you can ask anything you want provide any perspectives you're always welcome to join that way and then something that's incredibly helpful for the lab is to commit assuredly ahead of time and contribute to the dot zero making the slides and helping contextualize the paper ahead of the dot one and the dot two so we in the last few months have talked about in number 35 we talked about a tale of two architectures free energy its models and modularity check out the live streams to learn more i won't even attempt to summarize the papers other than just to state their titles in 36 we discussed modeling ourselves what the free energy principle reveals about our implicit notions of representation we then dove a bit into the technical in 37 with free energy a user's guide in 38 discussed the evolution of brain architectures for predictive coding and in 39 discussed morphogenesis as bayesian inference a variational approach to pattern formation and control in complex biological systems and then in 4d we discussed a free energy principle for generic quantum systems which was as anybody who was there before or during or after it happened can a test was quite a special and interesting experience and just to show what that coda page looks like if you head over there you'll see the upcoming streams here we are in the end of march it's not in the live stream section it's in the um round table section here we are right now but in live streams you'll see the papers that we're going to be discussing in the coming months so if you're interested in any of these upcoming papers please get involved in the dot zero or the dot one and two if you're interested in inviting anybody or yourself being a guest on a guest stream that's always an affordance these are like one off small series that can be on active inference research or related topics and we've had some great discussions in the last few weeks on guest streams we have model streams and math streams that are for presenting some of the formal and model based components of active inference and org streams where we talk about organizational principles and organizational design so those are just some of the live streams that we've had and it's been fun especially if you want to give a guest stream or if you can connect to somebody who does want to or encourage or recommend somebody or you want to be a facilitator for someone else to present or any other combination of affordances let's make it happen because we can get a lot of awesome modifications in our niche i think what they used to call making content back in the web two days so that's all for live streams podcast blue do you want to give an update sure so we are moving through the podcast really revisits a lot of the early live streams so like if you're playing catch up like if you caught a live stream you're like i should go through the back catalog like maybe now you're on live stream 20 or something like uh the podcast starts with the early episodes and i think we're through episode nine live stream number nine now um so yeah check it out it's fun um also a great chance to get involved if you have like av expertise would love to get some pointers also like i know some stuff um i've been clipping the podcast for a little bit so if you'd like to learn like the basic things that i know and want to get involved in making the podcast reach out um i could always use help if you want to listen to more podcasts come help me make them dave yeah if you have uh listed the podcast after nine that you're planning to do i'd like to get that because i want to get started with uh just doing basic text conversion on this if that assumes that this these are the priorities is that correct these are the ones that you that we want to get in front of people i don't i don't know what the priority is there dave but i i will say like there's definitely value to having a transcript um uh i actually looked at podbean who's our podcast platform the other day and they will transcribe our podcast for us actually at 20 cents a minute so if you really want to know what the value of transcription is just just the value to just put it into text now i don't know if that comes with with periods or punctuation or i don't know what that output looks like but transcript there i'm not sure that these are necessarily the priority but it's just going through the back catalog it's helping me catch up so i just kind of started where i wanted to start at the beginning why not start at the beginning right i don't know what what i love about these podcasts is they're short they're like from four to maybe 15 or 10 minutes long the first time that a speaker speaks they introduce themselves and there have been some probable clippings and rearrangements which are very subtle that help that listenable piece be like coherent like it's like an extended thought even one that's distributed across multiple speakers so like you hit play listen to it and then you can think and reflect because sometimes when it's two hours on a paper it feels like it's like a rail car coming through it's like every single idea and thing and people say that they pause the live streams and there's so much and then also as the priorities um that's something we can figure out and discuss but won't it be amazing when the live stream is you know the front car of that train piercing into the future doing it live there's transcripts for all the live streams and that can be on one hand connected to the ontology a transcript of the live stream itself could exist there could be some clipping and some transcripts for the podcast so as we develop our informational ecology all of these kinds of connections and tasks are in play it's just a question of how late can you stay up which i've wondered about for a long time dave needs no sleep um you know the other cool thing about the podcast is that it's slide free so where when you're on the youtube watching the live stream often it's associated with some visual image and sometimes there will be like a description of the image in the podcast but it's nothing that you need to like pull over and look at which is sometimes if you're trying to like watch the youtube videos on the long drive it's like ah that might not be safe but the podcasts are definitely um you know commute friendly uh because you don't have to see anything you can just listen so that's another it just offloads the visual input as well good for the treadmill stuff like that great point on to eight so some updates on our um i'll just call it social media we have our twitter with um 1359 followers we have about i think we crossed 300 members in the discord um over 770 subscribers on youtube and some several hundreds of people reached in facebook anyone who is interested in automating social media that would be an excellent affordance blue yeah definitely i was just you took the words right out of my mouth and facebook uh is very new so we have like just 40 likes on our facebook page so if you're listening to this and you use facebook regularly for whatever i mean it will ping you to remind you about podcasts and sometimes live streams sometimes i say because like i said we need an automator so that's your expertise and you want to always be reminded about the live streams uh get in touch yes just to be able to preload a buffer here in the front car of the train it's sometimes hard to clean up the pieces and so if you're feeling like that's something that's interesting to you then there's many opportunities to have engagements on different platforms and just feel free to just engage in the comments and in the threads however makes sense great enough with social media um okay on to nine so now we're heading more towards the tools part of this discussion primarily the work of jakobs meckle has been just excellent on active block prints which is a package that we're developing in order to facilitate cognitive agent modeling and if you look up active block fronts on github you'll find the open source repository what we've done with active block prints is implement active inference entities the kind that you will see graphically described in papers or the kind that have been implemented in python via pi mdp and we built upon pi mdp as well as a complex systems modeling framework called cad cad by integrating the active kernel of the entity action perception loop with the complex systems modeling framework we hope to have engineering-grade approaches towards complex cognitive ecosystems so if you are interested in python or julia or other aspects that are computational related to active inference and you like complexity and complex adaptive systems everything else that you might see on this slide active block for instance is an open source project so go ahead and fork it make your contributions and come get in touch with us if you'd like to be a little bit more in the loop really looking forward to seeing how this project develops in the coming years in dot tools also we've been having some awesome discussions around robotics and embodied cognitive entities so we have been characterizing kind of like a literature review on different cognitive agent modeling frameworks so active inference is often proposed as a unified cognitive entity model and so we've been just curating many other cognitive agent modeling frameworks from the past and present and then assessing them in terms of their language what cognitive features they implement whether they deal with embodiment what is their mathematical underpinnings and so on and meanwhile characterizing some of those cognitive features and elements like anticipation memory counter factuals abductive reasoning different kinds of cognitive features or motifs and then trying to understand how do these different cognitive model frameworks line up on the rubric of wanting to have expressivity across all of these different cognitive features and to integrate them and that's a little bit more on the um symbolic cognitive we've also been thinking a little bit about the embodied whether the embodied and spatial and bodily or the embodied and robotics so here is um a screenshot from work of jf cloudshire who's been an awesome participant in this past quarter and some active robots that are uh carl and andy i think people can probably guess what their last names are there isn't too much more to add here other than um check out this woefully under acknowledged channel and work and if you're interested in any of these areas like related to cognitive and robotics and embodied cognitive systems those are the kinds of things that will be awesome to scaffold in the and to our final of the 11 projects actin flab meta modeling so uh actin flab meta modeling aka alma so there's an a and an l and there's an m and there's an a and i think that was um jessica's suggestion because alma has nice connotations as well this project is using meta modeling tools like coda to provide the capacity for lab scale generative modeling updates and reconfiguring so the ability for the lab to reflect and to modify itself and its niche in order to design the kind of participatory niche that facilitates successful policy selection by lab members like what does the lab need to provide the individual as the action state of the lab sense state of the individual in terms of the communication such that that participant is chill but also making the right decisions that are productive we're using primarily coda that's why the font is bigger here but we're also implementing the active entity ontology for science that we discussed on the research slide and active block for ins just two slides ago as a more formal way to get at some of these meta modeling and this is a figure from our 2020 paper that first early paper on systems engineering and active inference and remote teams where we also looked at the informational niche the shared informational niche of the team this is their shared folder their shared code document their formal documents their drive the chat threads that's the shared epistemic niche where actions to modify the niche by one participant can then be like stigma g modifying the environment so that themselves later or other participants then or in the future take different action selection as mediated by the way that that sensory state updates their generative model and then we have narrative and so we've been very interested for a long time in rhetorical and narrative approaches to cognition and so now we're just beginning to see how some of these approaches can come into play in the context of active inference and in the so that's like the meta on actin flab and this is also an project set of affordances if you like coda if you liked douglas hofstetter's girdle escherbach if you like act infer serve all of these would be great diagnostic cues that perhaps you would like active lab meta modeling yes steven yeah thanks for sharing i was wondering how this also might even be able to tie in in the future to uh micro phenomenological kind of modeling i that that kind of approach so we've got the meta the narrative the story and uh hopefully maybe some sort of approaches as well but i think we've got to walk but we'll get there so thanks when you said micro phenomenological i thought oh like the experiences of children say hello um cool so that is it for the project updates they're all ongoing let's just kind of go over some several areas that we're especially looking for assistance or contributions on would be on the legal side um for example potentially connecting with a firm that would be interested in providing pro bono assistance who has experience with education nonprofits intellectual property tech etc and also helping our lab navigate a course towards financial sustainability whether through grants or other mechanisms but just finding a way that we can reward and stabilize the contributions of participants and then at the project scale any and all participants who want to be in the strange loop in the strange attractor want to be on the path towards contributing to active inference lab as they also grow and develop as an active inferencer and as a person so if any of these projects or some other opportunity that you see and want to take the initiative on any of these reflect affordances for you so hopefully if you're listening this far in a minute and six hours in or wherever it's clipped out in a podcast then just recognize that these are all reflecting the past contributions of like yourself and this is just an opening it's only the first quarter of 2022 we're just beginning and so there's so much in the field and in the lab to be done that there will be something to do that's meaningful and chill and aligned with your preferences and expectations and availability just believe it so that takes us close to the end um and a little modification of our common ending slide where do we go from here what are people still curious about learning and applying and what are we going to do differently maybe in the next quarter yes blue so i think i already put it out there at the beginning um but i'm really like interested in not talking about things anymore spending less time talking about things and more time actually doing things so that that's what i'm going to do differently in the next quarter and that's me as like on a personal scale has nothing to do really with the lab but it's i just want to just work now i just want to put my head in the books and get to it awesome yeah sometimes like it helps to get caught up through conversation and just being in the synchronous meetings but one hour per week is not how we build a course or not how we go through a textbook and so that is a very important balance to keep in mind stephen or dave any other thoughts or no there's a lot of good work going on i'd say just keep doing it and talk about you know talk about what's working well what needs to be improved what the bottlenecks are we've mentioned some bottlenecks and know sometimes uh maybe some of the fun projects that could be pushed back in favor of some of the more difficult things that actually will end up being leveraged heavily the i think of the ontology immediately it may not be fun to slog through this stuff but it's going to really pay off when you get some transitive closure aspects of it of course awesome for those who are studying or curious about active inference but feel kind of like on their own whether no one else in their local niche or network or lab or university or organization is working on active inference that's like one of the most powerful things is just knowing like if you have an idea that you're not even going to pursue but just somebody should do this or it would be cool if this existed having a space where you can add that pebble and maybe down the road or maybe sooner than you expect somebody else seeing that pebble is stimulated to act and so that's what can happen when we have shared ways of working and that's just what we're beginning to see stephen yeah sort of bouncing off that as well i think i keep thinking about stages of readiness which i i am i'm normally working with community participants who are marginalized but i kind of i've had that myself with like feeling a bit marginalized feeling am i like ready and there was like a little jump but it can be a chasm you know trying to get to that stage and i think that can be for a lot of people you don't quite know on different things whether you're near that threshold or not and you're often a lot closer to being ready and you maybe just need someone to give you a kick up the bum and say just go for it you know but having said that that's when you appreciate the quality of getting get getting some things under your belt on the way to the line so i think uh like blue said like it's time to act and um for me personally yes like um the stage of readiness i'm at is very different to where i was a year ago for sure both in my knowledge when i look back to where i was but also in my heart and that's a that's not an insignificant piece of the equation so thanks awesome i'll just wait one more minute if anyone who's watching live has any questions that they would like to see addressed but totally agree it's um it's a multi-year journey i mean we're all on a lifelong journey of some sort but hopefully even our active inference journey is multi-year being able to connect our policy selection our behavioral decisions in the next micro cicade of the eyes the next place your eye is going to look the next action you're going to take in terms of opening up a hyperlink or reaching for a book or a pencil but being able to connect those very rapid actions with a bigger picture and with the longer temporal time scale and acting amidst uncertainty that's what active inference is one aspect of it and so if anything that you've heard resonates just hopefully see this round table presentation as like an invitation less than a uh stand and deliver because there's so much openness in the area and in the lab and maybe can feel like one is starting late but that's just human knowledge development it's a treadmill that's been going and through coming together in the lab we can actually leverage each other's contributions and catch up on fields that have thousands of hours of reading hopefully get somewhere cool together so dave blue steven and yvonne and alex from earlier and all the other lab participants big thanks for this amazing quarter and it's been a fast one on to the next one bye thanks all have fun you